Glaze Search · step-by-step guide

Card G finds Glazy glaze recipes from what you have: **your own words**, **a colour**, **a photo of a pot or test tile**, or **a glaze formula**.
It runs on your PC (http://127.0.0.1:8770); only the recipe photos come from Glazy's website.

What is new NEW

- **Describe it**: type freely - typos, shorthand (`^6`, `c10`, `ox`) and filler words are fine. The page shows the keywords and values it understood.
- **Smart search**: one model that compares your text, a photo object and a formula with every recipe, plus colour, cone and surface.
- Colour is matched against **all colours** of a recipe's photo (two-colour tiles count).
- Cone **04 is not cone 4**: low fire stays low fire.

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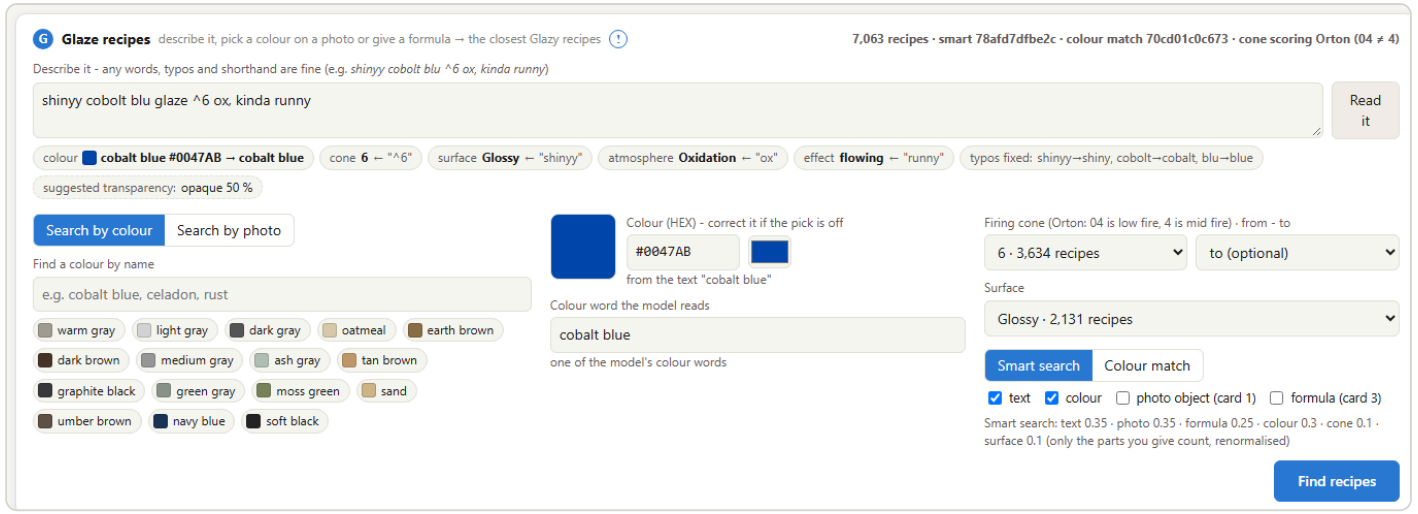
The six steps of a search (NEW = added by the smart-search upgrade).

0 Start the app

1. Only once, after `SETUP.bat` and `DOWNLOAD_DATA.bat`: drag the model ZIP (`test_similarity.zip`) onto **INSTALL_GLAZE_MODEL.bat**. It installs the colour-match model and trains the smart-search model (about 2 minutes with an NVIDIA GPU).
2. Double-click **RUN_GLAZE_SEARCH.bat**. Wait for `Glaze search is ready` in the black window; the page opens by itself.
3. Scroll to **card G · Glaze recipes** (below card 1b). The top line shows the number of recipes and both model versions.

> Keep the black window open while you search. Closing it stops the app.

> If card G says *the model is not available*, run INSTALL_GLAZE_MODEL.bat and restart RUN_GLAZE_SEARCH.bat.



1 Upload a photo

In **card 1 · Photo**: drop a photo, press Ctrl + V to paste a screenshot, click to browse, or paste an image link.

The app finds every ceramic object (Grounding DINO), draws its exact outline (SAM 2.1) and checks that it is ceramic (CLIP). Each object gets a number and a colour outline.

Example: a photo with **four test tiles** - pink, orange, brown and yellow glaze, each with bare clay at the bottom.

> If the photo is already in the database, a grey note says so - that is fine, the steps are the same.



2 Select the object you mean

Keep the photo in **Select objects** mode.

- **Hover** an object: it lights up and shows its number and how sure the app is that it is ceramic.
- **Click** it: it becomes the **active object** (thick yellow outline).
- Ctrl +click adds or removes more objects.

Here the **yellow tile (#4)** is selected. Card 1b now shows *this* object: its colour regions and attributes.

> Choose the object whose glaze you want - not the one that fills most of the photo.



3A Extract the glaze colour - pinpoint it

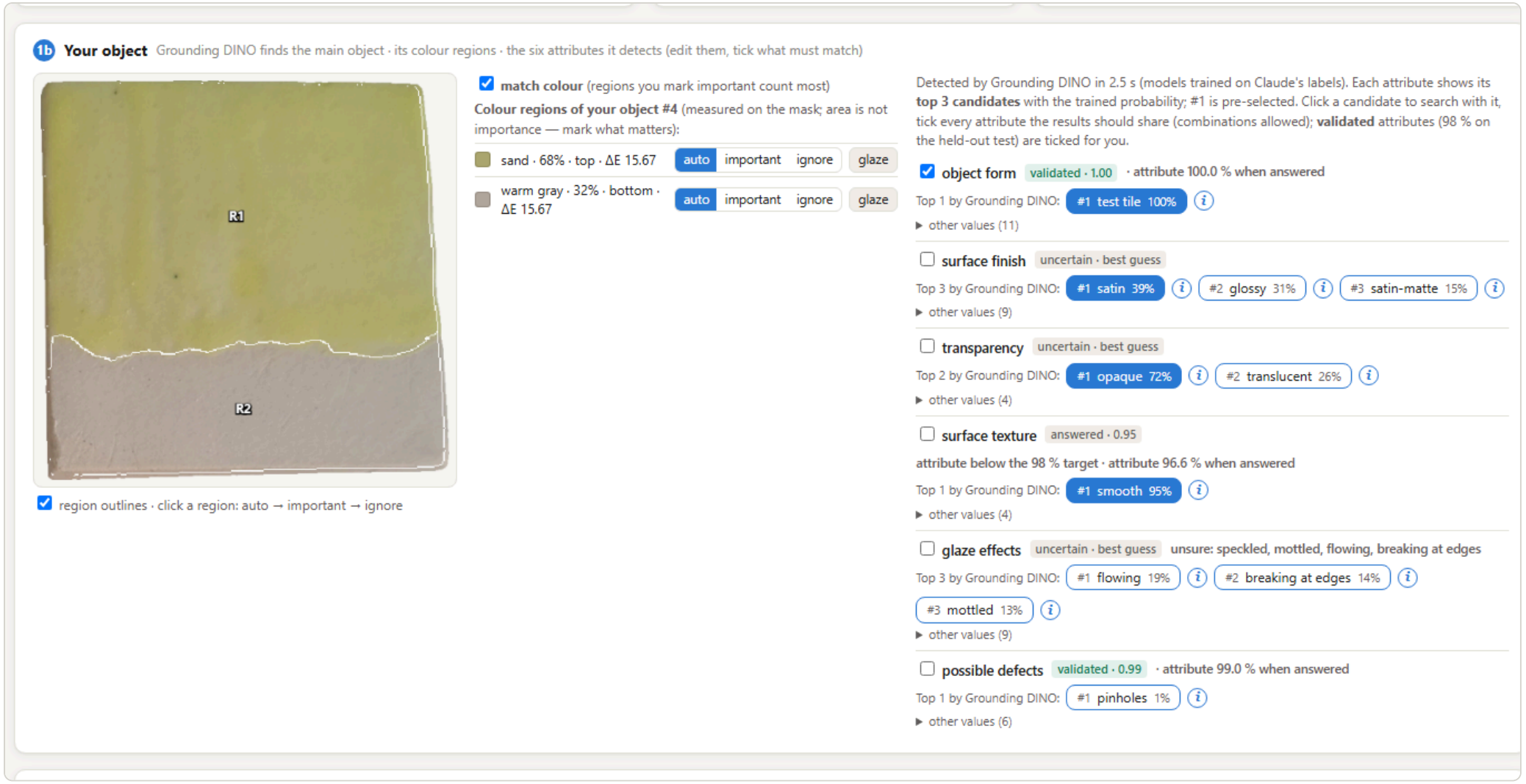
1. Switch the photo to **Pick colour (HEX)**.
2. Click on the **glaze** - not on bare clay, a shadow or a reflection.
3. The colour is the median of 13 × 13 pixels **inside the object's outline** (the background is never used); the brightest and darkest 15 % are left out, so one reflection cannot shift it.
4. The box under the photo shows the colour and its HEX. **Correct the HEX by hand** if it looks off.
5. Press **Use for glaze search**: the colour goes to card G, which switches to *Search by photo*.

> Here: a click on the yellow glaze of tile #4 gives `#B2A962` (median of 119 pixels).



3B Or choose a colour region of the object

Card 1b splits the active object into its colour regions (outlined and numbered R1, R2 … on the picture). This helps when a pot has several colours or bare clay. On this tile: **R1 = the yellow glaze** and **R2 = the bare clay foot**. Press **glaze** next to R1 to search with the glaze colour only.



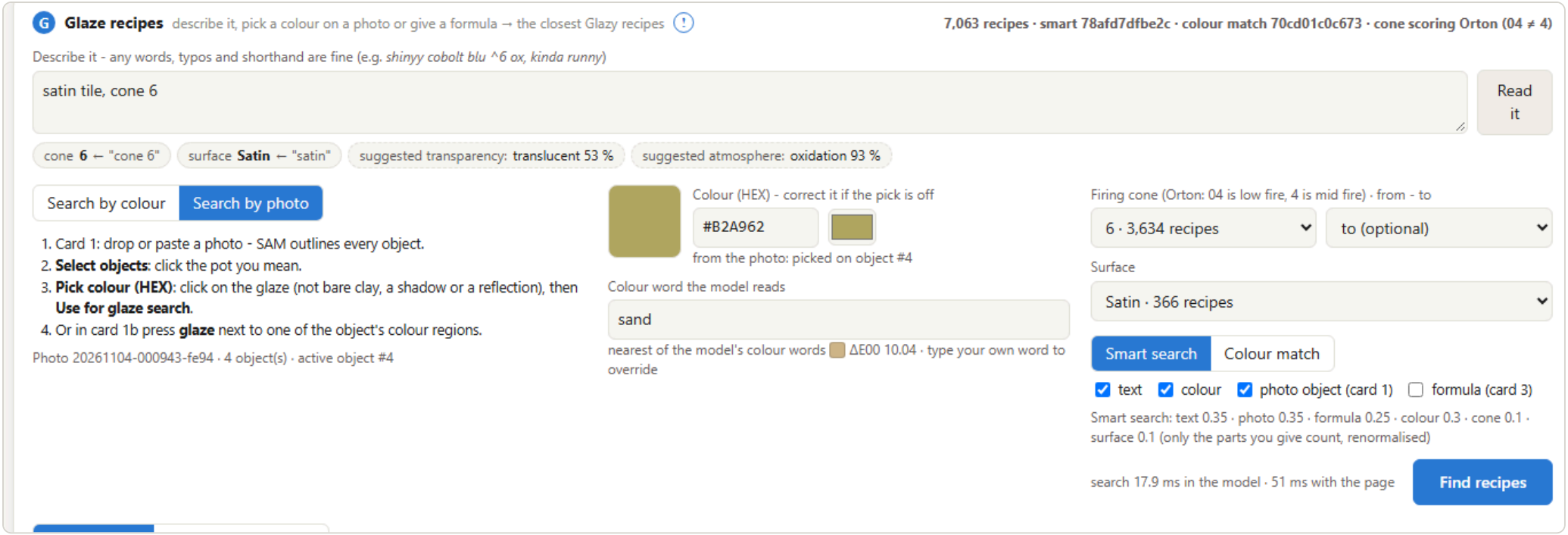
4 Describe it in your own words → keywords and values

Type in **Describe it** (top of card G) and press **Read it** (or just **Find recipes**). The chips show what was understood and **which of your words** it came from; the colour, cone and surface fields are filled in for you. Dashed chips are *suggestions*: tags your words imply, learnt from 4,766 recipe descriptions on Glazy - click one to use it.

What you type (noisy on purpose)	Colour → HEX → model word	Cone	Surface	Transp.	Other values · typos fixed · suggestions
`shinyy cobolt blu glaze ^6 ox, kinda runny`	cobalt blue `#0047AB` → *cobalt blue*	6	Glossy	-	atmosphere: Oxidation effects: flowing typos fixed: shinyy → shiny, cobolt → cobalt, blu → blue *suggested: transparency opaque 50 %*
`looking for a satin matte light celadon cone 10 reduction pls`	light celadon `#DAFFDB` → *mint green*	10	Satin-matte	-	atmosphere: Reduction *suggested: transparency translucent 70 %*
`low fire red earthenware, opaque, cone 04 with crackle`	red `#FF0000` → *scarlet red*	04	-	Opaque	effects: crackle *suggested: atmosphere oxidation 56 %*
`glossy black tenmoku c10 reducton with 8% red iron oxide`	black `#1B2633` → *blue black*	10	Glossy	-	atmosphere: Reduction colourants: red iron oxide 8.0 % typos fixed: reducton → reduction *suggested: transparency opaque 55 %*
`matt white liner for my mugs mid fire`	white `#FFFFFF` → *porcelain white*	5 - 6	Matte	-	*suggested: transparency opaque 72 %, atmosphere oxidation 90 %*
`SiO2 64.2 Al2O3 13.1 CaO 9.0 K2O 3.2 Fe2O3 2.0 cone 6 glosy`	-	6	Glossy	-	formula: SiO2 64.2 · Al2O3 13.1 · CaO 9.0 · K2O 3.2 · Fe2O3 2.0 typos fixed: glosy → glossy *suggested: atmosphere oxidation 76 %*
`Light Bluish Gray SEMI MATT cone 6-7 oxidation`	light blue gray `#92A2AB` → *cool gray*	6 - 7	Semi-matte	-	atmosphere: Oxidation typos fixed: bluish → blue *suggested: transparency opaque 55 %*
`dark greenish speckled stony glaze cone6 wood fired`	dark greenish `#005C00` → *forest green*	6	Stony Matte	-	atmosphere: Wood effects: speckled *suggested: transparency opaque 71 %*
`a celadon for my gas kiln`	celadon `#ACE1AF` → *mint green*	-	-	-	*suggested: surface glossy 51 %, transparency translucent 70 %, atmosphere reduction 81 %*

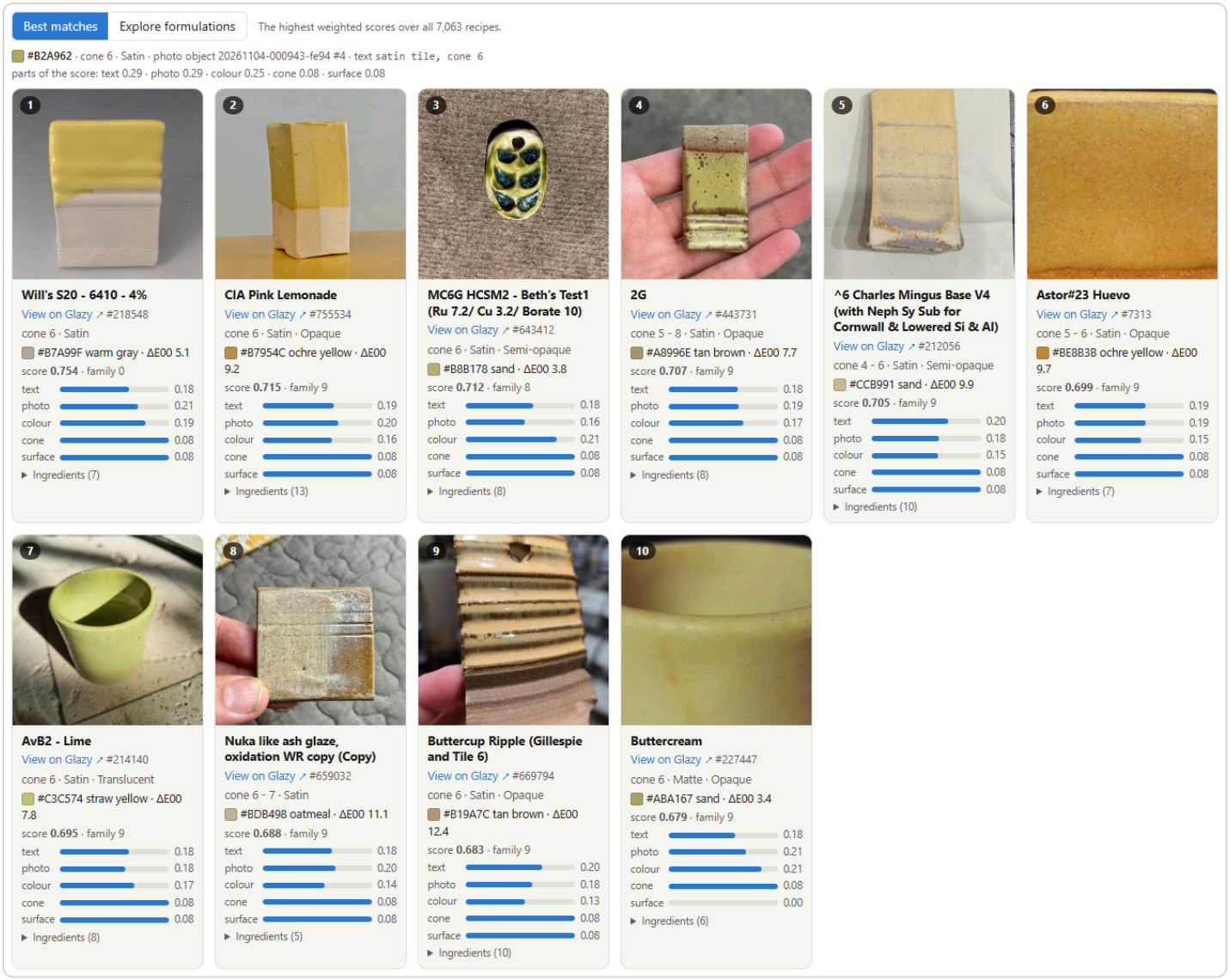
Real outputs of the app. Tested on 1,059 noisy sentences of held-out recipes (15 % typos): when it reads a value, it is right for colour 93 %, cone 99 %, surface 89 %, atmosphere 100 %. Words it does not know are still read by the text model.

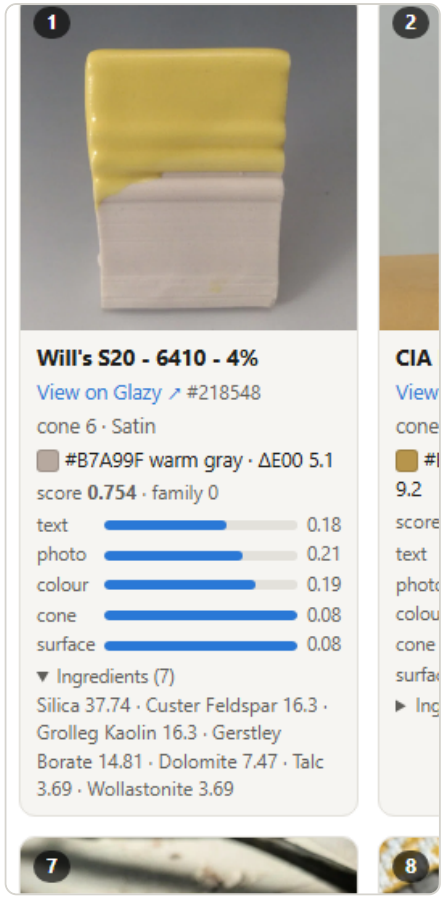
5 Choose what to use and search



- **Smart search** (default) uses everything you tick: **text**, **colour**, **photo object** (the active object of card 1), **formula** (card 3), plus cone and surface.
- **Colour match** uses one colour + cone + surface only (the earlier text-chemistry model).
- Here: the yellow colour picked on tile #4, the tile itself as photo, and the text `satin tile, cone 6` (surface and cone read from it).
- Press **Find recipes** (or Enter). One search takes about 0.1 s.

6 Read the results

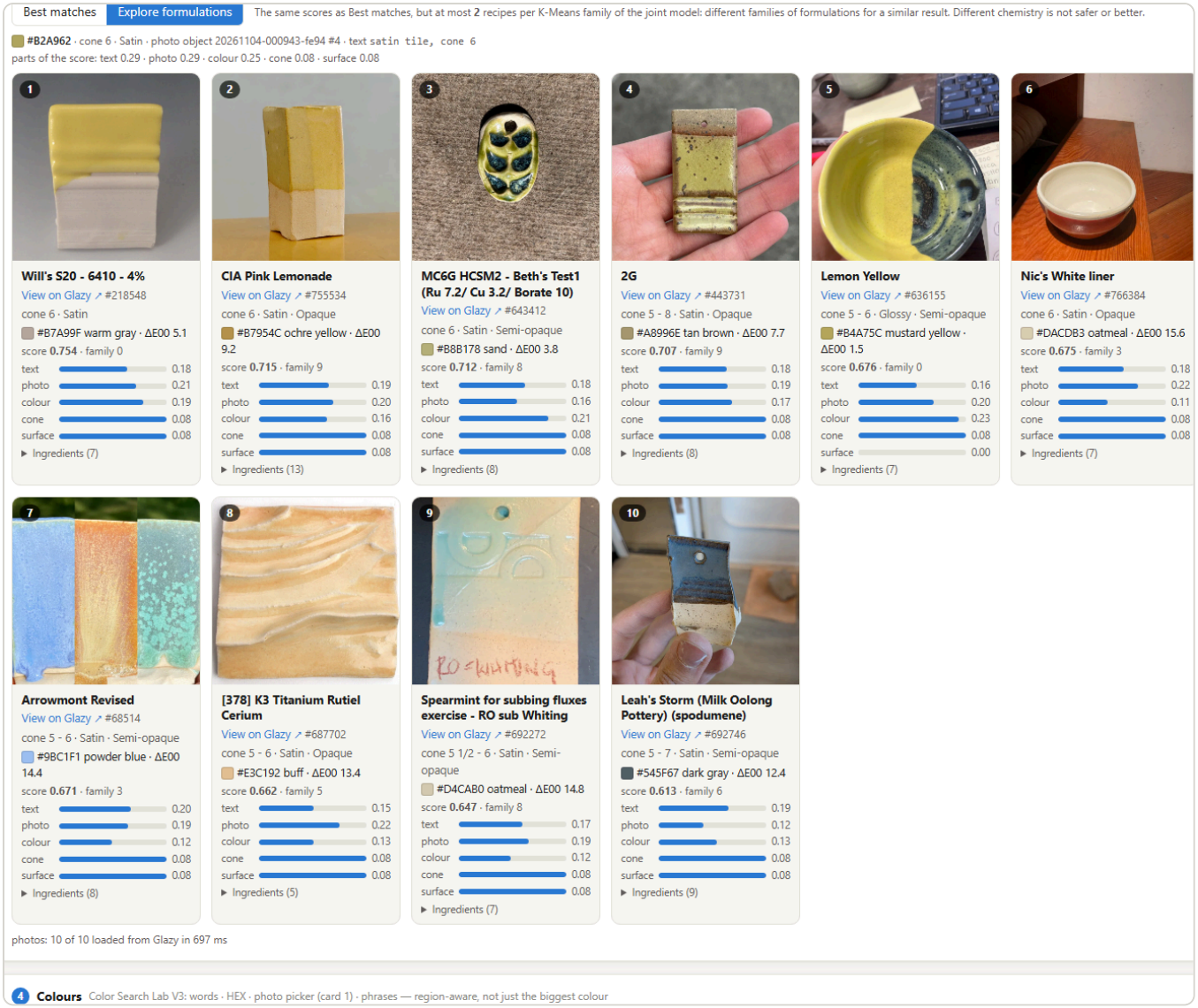




- **Photo** from Glazy (click: full size)
- **View on Glazy**: the recipe page
- cone · surface · transparency
- recipe colour and ΔE00 to yours (lower = closer)
- **score** and its parts (text, photo, colour, …)
- ingredients (click to open)

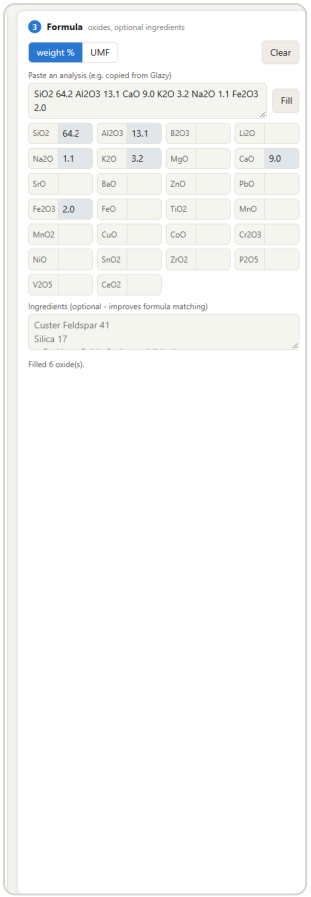
7 Explore formulations

The second tab uses **the same scores**, but takes at most 2 recipes from each family of similar recipes, so you see different ways to make a similar glaze. Switching tabs does not search again. Different chemistry does not mean safer or better - check every recipe before you mix it.



8 Search by a formula · tips

Paste an oxide analysis in **card 3** (e.g. copied from Glazy), press **Fill**, tick **formula (card 3)** in card G and search. You can also write the oxides in the text (`SiO2 64.2 Al2O3 13.1 CaO 9 …`).



Tips

If …	do this
the picked colour looks wrong	click again on clean glaze, or type the HEX; or use a colour region (step 3B)
results have the wrong cone	check the cone chips / dropdown: **04** is low fire, **4** is mid fire
a word is not understood	it is still read by the text model; add a colour, cone or surface word to make it explicit
photos show "photo unavailable"	the photos come from Glazy: check the internet connection (the search itself works offline)
card G says the model is not available	run INSTALL_GLAZE_MODEL.bat, then restart RUN_GLAZE_SEARCH.bat

How well it works (held-out recipes)

- Noisy sentence → the right recipe in the top 10: 74 % (pretrained text model alone: 62 %).
- Photo → recipe top 10: 45 %; formula → recipe: 33 %; random: 1 %.
- Top 10 that match colour, surface and cone of a noisy sentence: 73 % (text model alone 54 %, read values alone 55 %).